

# Sahara Mahbuba

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Country: Bangladesh

Civil engineering graduate with a strong interest in oceanography, particularly in coastal and marine environmental systems. Interested in the transport and transformation of carbon, nutrients, and contaminants within aquatic environments, with a focus on biogeochemical processes and water quality. Trained in numerical methods and computational tools for analyzing complex systems, with a quantitative and interdisciplinary approach to environmental research.

## EDUCATION

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### Bachelor of Science in Civil Engineering

Bangladesh University of Engineering and Technology (BUET), Feb 2026

CGPA: 3.14 / 4.00

## ACADEMIC INTERESTS

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| - Coastal and marine environmental processes | - Water quality and treatment |
| - Marine geochemistry                        | - Wastewater management       |
| - Contaminant dynamics                       | - Microplastic pollution      |

## TECHNICAL SKILLS

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General Purpose Software: Microsoft Word, PowerPoint, Excel

Language: Python, C++

Programming and Numeric Computing Platform: MATLAB

Drafting Software: AutoCAD

Structural Analysis and Design Software: ETABS, SAP2000

FEM Software: Abaqus

Statistical Software: RStudio

Methods: Finite element analysis (FEM), numerical methods, basic data analysis

## ACADEMIC WORKS

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- **Capstone Project: Multipurpose Shopping Complex, Rajdhani Supermarket, Tikatuli**  
Achieved comprehensive environmental impact assessment for the capstone project, measured by systematic identification of environmental risks and mitigation considerations, by independently conducting a literature-based EIA using published reports, standards, and prior studies.
- **Curve Setting Calculator**  
Achieved automated roadway curve analysis, measured by validated computation of standard horizontal and vertical curve parameters from minimal input data, by designing and implementing a MATLAB application based on textbook methodologies.
- **Landscaping of Garden**  
Achieved nature-focused public space design, measured by integration of biodiversity-supporting elements that enhanced human-nature interaction, by developing the landscape design component of a collaborative, hypothetical garden project.

- **Environmental and Socioeconomic Effect of Matarbari Coal Power Plant Project**  
Conducted literature-based research on environmental and socioeconomic impacts, synthesizing data from project reports into an academic poster.

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## PROFESSIONAL AFFILIATION

Student Member, ASCE

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## THESIS

**Title:** A Parametric Study of Strength Reduction in Thin-Walled High-Strength Steel Plates with Bolt Holes Considering Curling Effects

**Supervisor:** Dr. Bashir Ahmed, Professor, Department of Civil Engineering, BUET

**Description:** Conducted finite element analysis (FEM) in Abaqus/CAE to explore the effects of plate thickness on curling behavior, stress distribution, and failure in bolted thin-walled carbon steel plates, highlighting simple performance patterns.

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## EXAMS & CERTIFICATIONS

GRE: Total 324 (Quant: 167, Verbal: 157, AWA: 3.0)

IELTS: Overall- 8.0/9.0 (Listening: 9.0, Reading: 8.0, Writing: 7.0, Speaking: 7.5)

Online Courses: MATLAB Onramp – MathWorks (Certificate obtained, 2022)

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## EXTRACURRICULAR ACTIVITIES

- General Member, BUET Film Society (BFS), 2021-22
- Publication Secretary, BUET Film Society (BFS), 2022-23
- General Member, Charcoal- BUET Artista Society, 2021-2023
- General Member, BUET Origami club, 2022-23

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## LANGUAGES

- English: Full Professional (CEFR C1)
- Bengali: Native
- Italian: Elementary
- Korean: Elementary

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## REFERENCES

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| <ul style="list-style-type: none"><li>• <b>Dr. Bashir Ahmed (Thesis Supervisor)</b><br/>Professor<br/>Department of Civil Engineering<br/>Bangladesh University of Engineering and Technology (BUET)<br/>Dhaka-1000, Bangladesh<br/>E-mail: <a href="mailto:bashir@ce.buet.ac.bd">bashir@ce.buet.ac.bd</a></li></ul> | <ul style="list-style-type: none"><li>• <b>Dr. Tahsin Reza Hossain</b><br/>Professor<br/>Department of Civil Engineering<br/>Bangladesh University of Engineering and Technology (BUET)<br/>Dhaka-1000, Bangladesh<br/>E-mail: <a href="mailto:tahsin@ce.buet.ac.bd">tahsin@ce.buet.ac.bd</a></li></ul> |
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